

HSE TOOLBOX TALK

GRINDING AND ABRASIVE WHEELS

PURPOSE AND SCOPE

Prevent wheel burst, kickback, sparks, noise and dust during grinding through correct wheel-machine matching, guarding, two-handed use, workpiece restraint and spark control.

KEY HAZARDS

1



WHEEL BURST

2



KICKBACK

3



UNGUARDED TOOL

4



WRONG SPEED

5



PINCHED WHEEL

6



SPARKS

7



DUST

8



NOISE

CRITICAL VALUES TO VERIFY IN THE FIELD

Wheel RPM ≥ tool

Marked speed compatibility

100%

Guard and side handle fitted

0

Cracked, dropped or unidentified wheel

REMEMBER - SERNEM

- S

Stop and think
- E

Evaluate the risk
- R

Remove or reduce
- N

Never assume
- E

Ensure protection
- M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1926.300 and 1926.303 address portable-tool guards and the selection, mounting, inspection and safe use of abrasive wheels.

OSHA 1926.300(b)

GUARDING

Guard moving parts and the point of operation; keep the manufacturer guard correctly positioned.

OSHA 1926.303(b)

WHEEL INSPECTION

Closely inspect before mounting; ring-test suitable vitrified wheels as required by the manufacturer.

OSHA 1926.303(c)

SPEED AND MOUNTING

Verify wheel rated speed is not below tool no-load speed and that mounting flanges are suitable.

ÜRETİCİ TALİMATI

USE AND LINE OF CUT

Use the correct wheel surface, avoid side loading and binding, and keep the body out of the wheel plane.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|---|
| ☐ Daily tool inspection | ☐ Wheel type / RPM marking |
| ☐ Guard and handle check | ☐ Electrical / cord check |
| ☐ Hot-work / spark assessment | ☐ Dust and noise control |

SUPERVISOR DISCUSSION POINTS

- Is the wheel RPM suitable for this tool?
- Which way will the tool kick if the wheel binds?
- Where do the sparks end up?

CORRECT GUARD, TWO HANDS AND RESTRAINED WORKPIECE



Wheel/RPM - Guard - Handle - Clamp - Spark screen - Eye+face

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. REDUCE CUTTING / GRINDING

Consider pre-sized material, a saw or a lower-energy method.

2. CORRECT WHEEL AND GUARD

Use an undamaged wheel suited to material and RPM, with correct guard and side handle.

3. RESTRAIN WORK AND USE TWO HANDS

Clamp the workpiece and feed under control with two hands after full speed is reached.

4. CONTROL SPARKS, DUST AND PPE

Screen sparks away from combustibles and use eye+face, hearing and suitable dust protection.

CRITICAL WARNING



Never remove the guard, fit an oversized wheel or side-load a cutting disc. Stop immediately for vibration or abnormal noise.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|--|
| ☐ Wheel suits task | ☐ RPM match correct |
| ☐ Wheel undamaged | ☐ Flange / nut suitable |
| ☐ Guard fitted | ☐ Side handle fitted |
| ☐ Workpiece restrained | ☐ Spark path clear |
| ☐ Eye+face protection worn | |

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ATTENDANCE AND APPROVAL

Project / Site:

Date:

Presented by:

Work area:

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			